

Pose-Guided Predictive Ballistics with Multi-Camera 3D Tracking

MSc Thesis Defense

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Motivation

Need for Low-Cost Pose-Reactive Ball Delivery

- Fixed launchers cannot react [Lobster Elite, Sec 1.1]

MoCap is costly and marker-based [OptiTrack/Vicon, Sec 2.1]

Training needs joint-specific delivery

Low-cost cameras can bridge the gap

This thesis builds pose-guided aiming



Figure 1. Soccer Innovations Ball Launcher



Figure 2. RIT Motion Capture Lab

Problem Statement & Objectives

Can a ball launcher use 3D body-joint tracking for safe autonomous aiming?

Objective 1

Ball localisation

Mean 3D ball error below 120 mm → Achieved: 95.17 mm corrected

Objective 2

Joint localisation

Mean 3D joint error below 180 mm → Achieved: 143.38 mm (62 trials)

Objective 3

Safe integration

Six-stage validation S0-S4 completed 2026-04-09. E-STOP <100 ms

Objective 4

Practicality

Commodity cameras, open-source stack, and about USD 200 perception hardware.

Takeaway: RQ1/RQ2 met within validated scope; RQ3/RQ4 satisfied for static + controlled live-aim only — moving-target firing is the unvalidated boundary (Sec 1.3, 6.4).

Background & Research Gap

Gap: low-cost systems rarely combine 3D joint tracking with physical launcher actuation.

Table 1. Existing categories and the gap this thesis fills

Approach	Cost	Accuracy	Markerless	Live-aim closed loop
Optical MoCap (OptiTrack, Vicon)	USD 50k–200k	Very High	No	No launcher
Commercial launchers	USD 200–4k	N/A	Yes	Open-loop only
Vision-guided sports robots	USD 2k–10k	Medium–high	Usually	Fixed zones
This work	~USD 200	Targeting-level	Yes	Aim + controlled single-shot validated

Takeaway: low-cost markerless 3D perception is connected to real launcher aiming; moving-target firing remains next.

Key Technical Components

Six contributions across perception, control, safety, and interaction

**01 | Pose-
reactive**

**02 | Validated
4-camera 3D**

**03 | High-
speed YOLO-**

**04 | Six-stage
safety**

**05 | Multi-
modal voice**

06 | Reproducible

Takeaway: a conventional launcher is upgraded into a pose-guided, safety-gated training system.

Proposed System — Pose-to-Aim Integration

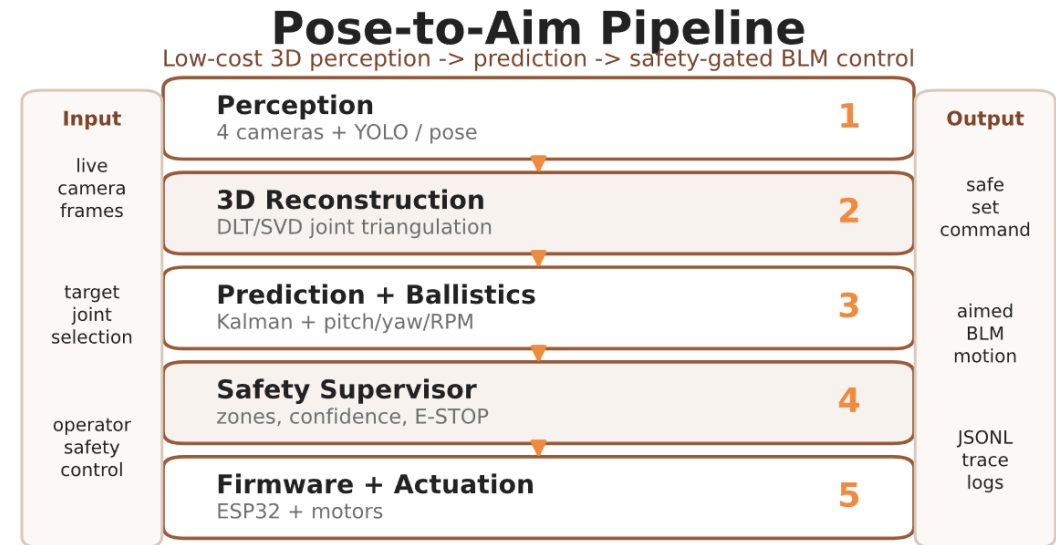
Upgrading inherited BLM hardware with vision-guided aiming, safety, and evaluation.

Inherited platform:
2-DOF BLM, flywheels, ESP32, pan/tilt

Thesis contribution:
4-camera perception, 3D triangulation, targeting

Core integration:
pose → prediction → ballistics → safety gate

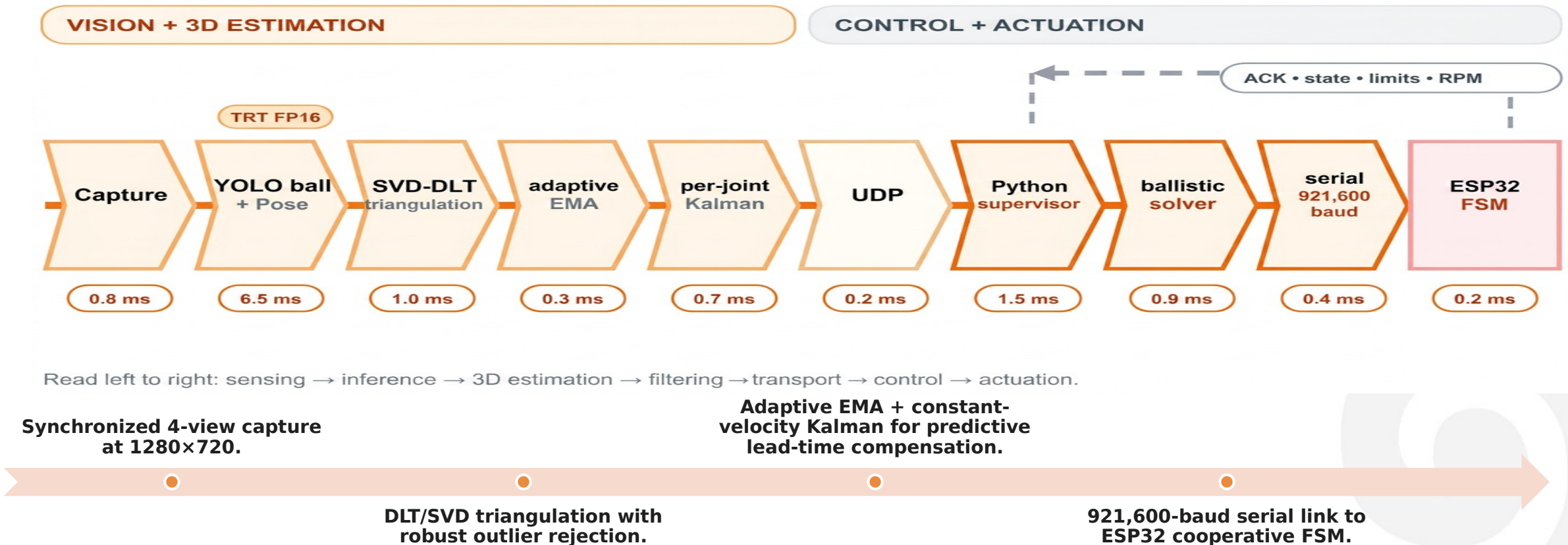
Validated scope:
static/live-aim tests; moving-target firing pending



Validated scope: static/live-aim integration. Moving-target closed-loop firing remains the next milestone.

Architecture & Pipeline

End-to-end runtime: ≈ 15 ms, below the 67 ms frame budget at 15 FPS.



Takeaway: compute pipeline ≈ 15 ms; 52 ms headroom remains within the 67 ms frame budget.

Hardware Implementation

Practical actuation paired with low-cost sensing — perception



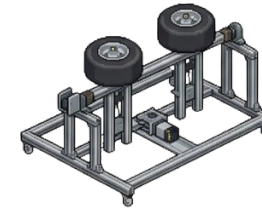
4× Hikvision DS-E12 USB cameras (1280×720, ~\$30 each).



ESP32 MCU running cooperative FSM (921,600 baud serial link).



NEMA-23 steppers with worm-gear reducers (self-locking for safety).



Arena setup: 4 USB cameras at corners, BLM at centre, 24 AprilTag fiducials on walls for extrinsic calibration (Sec 3.5). All hardware fits in a domestic garage — no lab infrastructure required.

Table 2. Indicative hardware cost

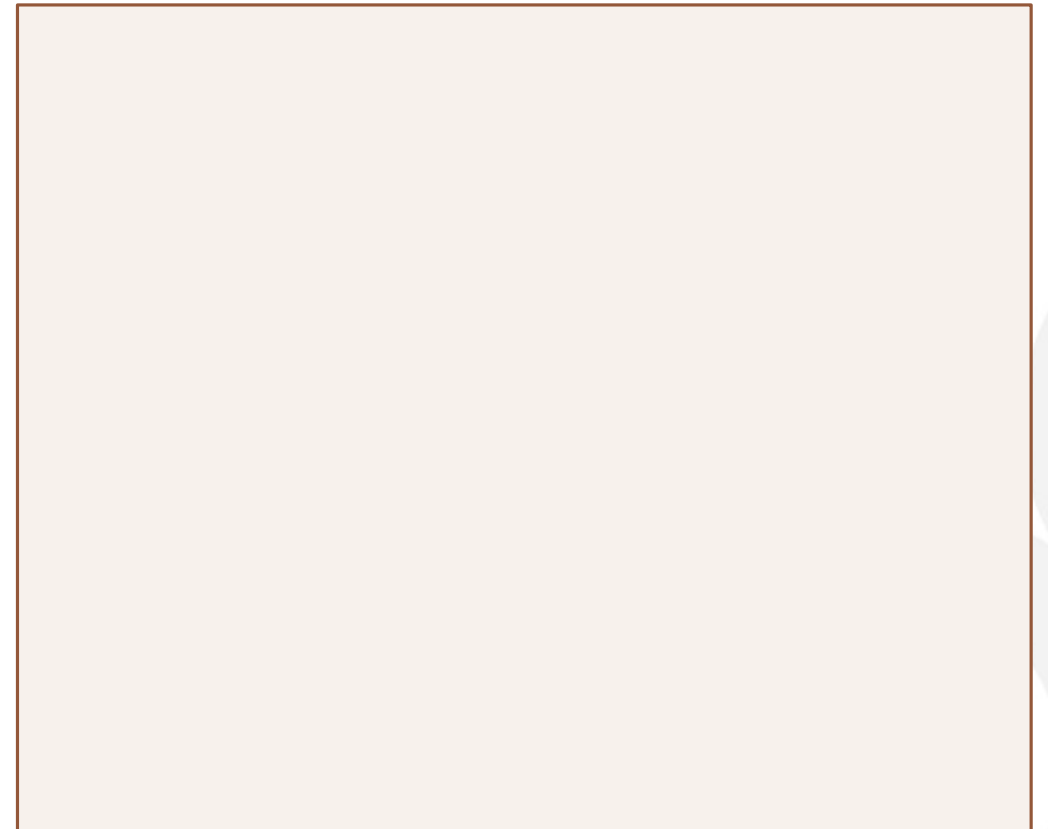
Item	Cost
Cameras (4×)	~USD 120
ESP32	~USD 8
Steppers + drivers	~USD 90
Flywheel motors + ESCs	~USD 60
Chassis + wiring	~USD 80
Total	≈USD 358

Takeaway: expensive MoCap is not required for pose-guided aiming; perception is ≈USD 120, with ≈USD 358 for the full BLM (Sec 3.2, Table 3.2).

Software Implementation

YOLO-Pose + TensorRT enables real-time 4-camera inference within the 67 ms frame budget.

Layer	Module / Tool
Perception	YOLO-Pose + custom YOLO ball detector
Inference	TensorRT FP16, 4-camera batch
Geometry	SVD-DLT multi-view triangulation
Robust ball	iterative reprojection-error rejection (Sec 3.7.2)
Filter	adaptive EMA + CV Kalman (Sec 5.7)
Supervisor	Python, safety gates, ballistic solver
Firmware	control_12_full.ino, ESP32 FSM



Takeaway: multi-camera detections become safety-gated pitch, yaw, and RPM commands.

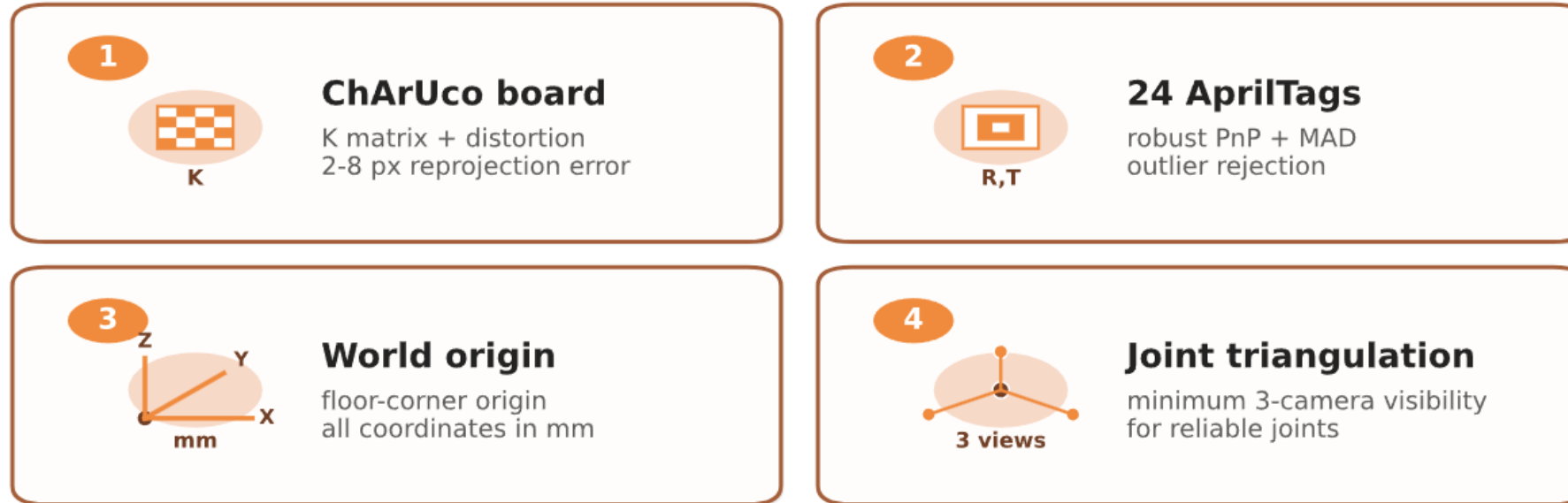
Calibration & Geometry

Intrinsic: ChArUco board, reproj 2-8 px (Sec 5.1)

Extrinsic: 24 AprilTags + RANSAC + $\sigma=2.0$ sigma-clipping (Sec 3.5.2)

Extrinsic RMSE: 3-7 px after 8-15 % outlier rejection (Sec 5.2)

Overlay va



Result: calibrated 2D observations become 3D world coordinates.

Takeaway: one calibrated world frame enables 3D joint coordinates from four USB cameras.

Methodology & Experiments

Evaluation combines localisation accuracy, dynamic tracking, safety, and traceability.

Experiment	Input	Metric	Outcome
Ball GT	36-point grid	Mean / P95	95.17 mm mean
Joint-touch GT	81 trials	Mean / P95	143.38 mm, 62 valid <800 mm frame-to-frame
Dynamic clips	3 clips	Stability	jumps (slow), motion-blur stress (fast), near-zero
Bias correction	Axis offsets from GT	Corrected vs raw error	Raw 150.77 mm → corrected 95.17 mm (Fig 5.1)
Safety checklist	S0–S4	Pass / fail + logs	Passed, E-STOP <100 ms
Decision logging	JSONL logs	Auditability	Every actuation logged

Takeaway: localisation, dynamic stability, and safety logging satisfy static + live-aim acceptance. Bias correction was fitted in-sample on the same 36-pt set (Sec 4.4.2, 6.3).

Key Results — Accuracy, Safety, Runtime

All thresholds met within validated static/live-aim scope. Raw ball mean 150.77 mm
→ corrected 95.17 mm via in-sample bias model (Fig 5.1).

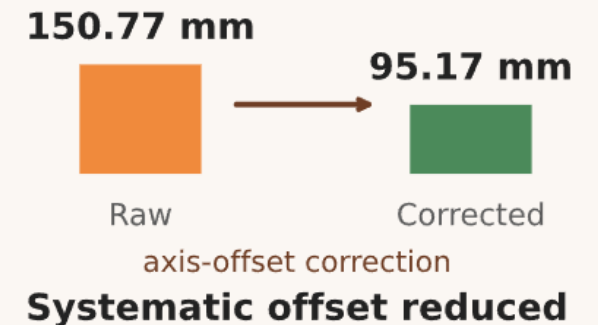
Headline results

95.17 mm	corrected ball mean (P95 166.51 mm)
143.38 mm	joint mean (P95 166.51 mm)
110 / 150 / 164 mm	knee / hip / shoulder means; P95 108.73 mm
<100 ms	171 / 172 / 170 ms which response (Table 5.3)
15 FPS	(Sec 3.14.3) pipeline, ~15 ms compute, 52 ms headroom (Sec 3.3)

Runtime comparison

YOLO-Pose		6.2 ms
MMPose		38.5 ms
Frame budget		67 ms
Fits the 67 ms frame budget		~6x faster

Bias analysis



Takeaway: localisation targets were met; safety and runtime support static/live-aim validation.

Limitations & Challenges

1. UNVALIDATED closed-loop firing at a moving subject (Sec 1.3, 6.3)
2. In-sample bias correction — fitted on same 36-pt set (Sec 4.4.2, 6.3)
3. 19 / 81 joint-touch trials invalid (23.5 %) due to occlusion (Sec 5.4)
4. RPM → velocity not empirically calibrated (Sec 5.9, 6.4.3)
5. Single-subject, indoor lighting, arena evaluation (Sec 6.3)



Boundary

Moving-target closed-loop firing pending



Generalisation

Single subject, one arena, indoor lighting



Visibility

19 / 81 joint trials invalid from occlusion



Ballistics

RPM → velocity calibration missing

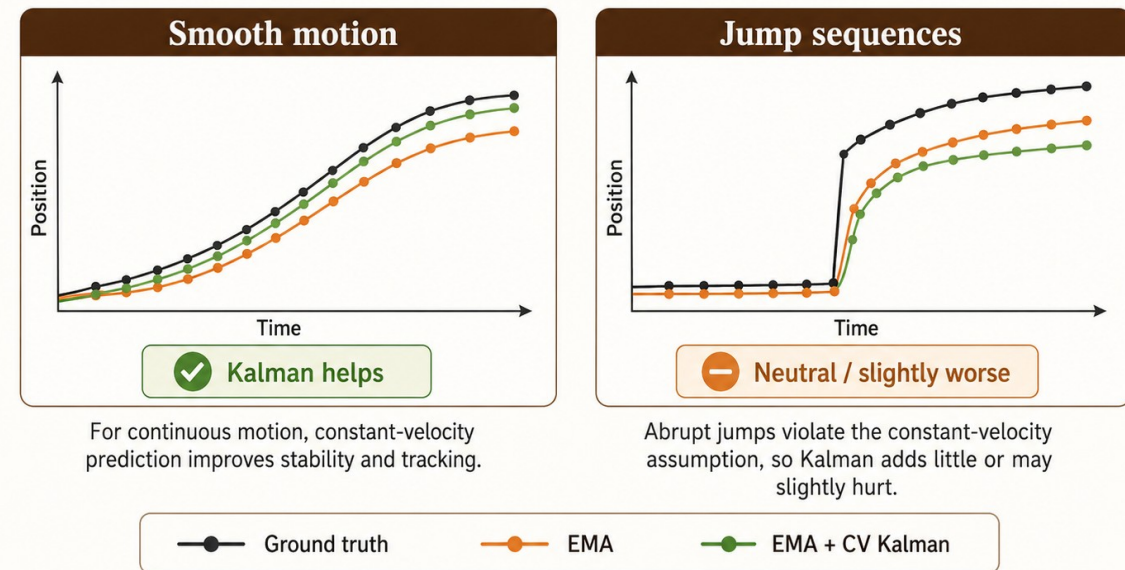


Hardware drift

Pan/tilt homing needs more validation

EMA Ablation: Smooth Motion vs. Jump Sequences

CV Kalman helps on smooth motion, but not on abrupt jumps.



Takeaway: validated scope (perception → safety-gated single-shot live-aim) is a strong partial validation. The unvalidated boundary — moving-subject closed-loop firing — is precisely defined and is the next milestone.

Conclusions & Contributions

Static/live-aim validation completed; moving-target firing remains future work.

1

Pose-reactive aiming

BLM aims live from 3D joints (Sec 5.9)

2

Low-cost 3D perception

4 USB cameras,
≈USD 120 perception / ≈USD

3

358 total
Safety-gated S0-S4 integration, E-STOP < 100ms
(Sec 3.14.30)
ms

4

Training-mode interface

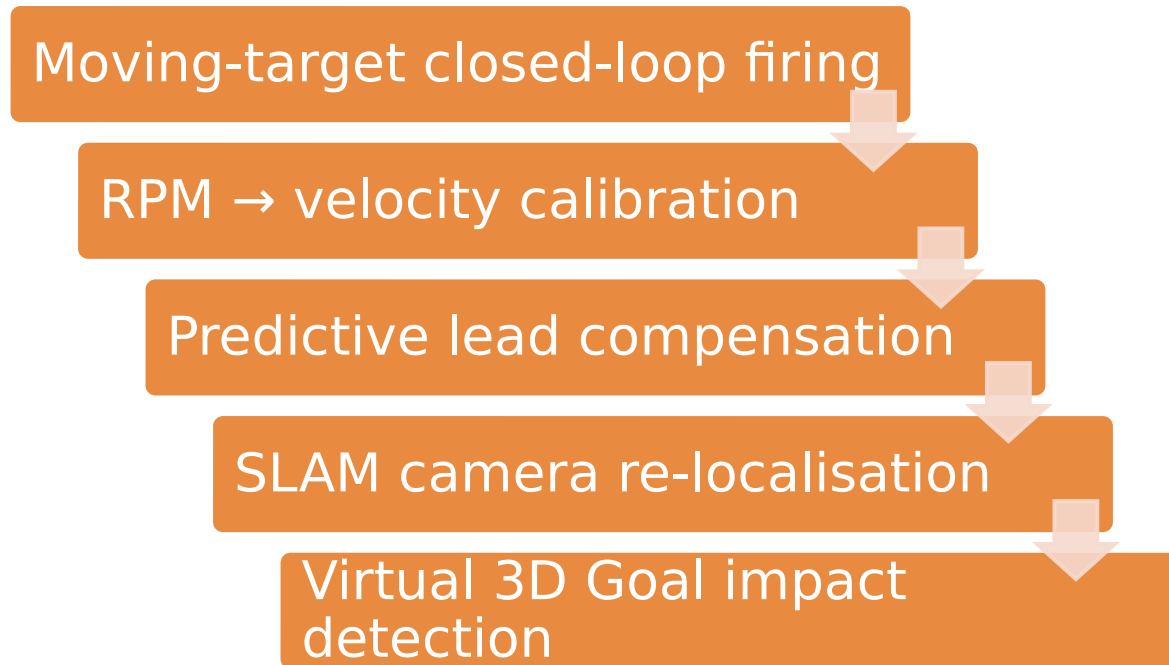
Voice + keyboard control behind safety gates (Sec 3.12)

Takeaway: strong partial validation of a low-cost, pose-guided BLM. Disadvantages: in-sample bias fit · 3-camera occlusion floor · RPM → velocity not yet calibrated.

Future Work + Ethics & Standards

Next milestone: moving-target closed-loop firing with calibrated RPM-to-velocity mapping.

Future roadmap



Applicability: sports training, rehabilitation, and adaptive PE instruction.

Dual-use: a system capable of autonomously tracking human body parts and directing a projectile has obvious dual-use potential beyond sports training. Operator presence, hardware NC E-STOP, exclusion zone, and the six-stage protocol are the necessary safeguards (Sec 6.5)

Safety & Ethics

ISO 12100

Machinery safety — L1–L10 hazard map (Sec 3.14)

ISO 13849-1

Safety-related control — NC E-STOP = ISO 13849-1 Cat-1 stop (L8)

IEC 60204-1

Wiring, fusing, E-STOP — 24V/50A fuse, single star-point ground (Sec 3.2)

ISO 10218-1

Operator-only zone during controlled fire (Sec 3.14.2)

Q: Why not use Intel RealSense depth cameras?

Cost, redundancy architecture, and occlusion tolerance drive the 4x USB camera choice

Cost comparison

- ▶ 4x Hikvision DS-E12 USB cameras: ~\$120 total (\$30 each)
- ▶ 4x Intel RealSense D435: ~\$800+ total (\$200+ each)
- ▶ Recycled PC handles all inference — no depth-specific compute needed

Redundancy architecture

- ▶ 4 overlapping views: if one camera is occluded, 3 remain for triangulation
- ▶ RealSense single unit = single point of failure; depth occlusion by ball or limb is common
- ▶ DLT/SVD triangulation explicitly designed for variable camera-count inputs

Accuracy argument

- ▶ 95.17 mm corrected ball mean achieved with commodity RGB cameras
- ▶ RealSense depth noise at 3-5 m range is 3-20 mm — marginal gain for 7x cost increase

Scope

- ▶ Controlled indoor arena only. Outdoor / variable lighting is acknowledged future work.

Answer: 4x USB cameras give occlusion redundancy, cost \$120 vs. \$800+ for RealSense, and already achieve 95.17 mm accuracy at indoor arena distances.

Q: How does the Kalman filter handle sudden position jumps?

Constant-velocity model with process noise — tested in Section 5.7, honest about limitations

Kalman model used

- ▶ Constant-velocity (CV) model: state = $[x, y, z, vx, vy, vz]$
- ▶ Process noise Q tuned to allow fast re-acquisition after direction changes
- ▶ Prediction window: 200-400 ms ahead to compensate mechanical settling time

Jump sequence behaviour (Section 5.7)

- ▶ On sudden position jumps, prediction error spikes then recovers within 2-3 frames (133-200 ms)
- ▶ Ablation showed neutral-to-negative improvement on jump sequences vs. EMA alone
- ▶ This is acknowledged honestly: Kalman helps smooth normal motion, not abrupt jumps

Why it still matters

- ▶ For steady-state walking/running, CV Kalman reduces targeting jitter significantly
- ▶ The 200-400 ms prediction covers mechanical lag — without it, the shot always hits where the target WAS
- ▶ Extended Kalman or IMM models are future work for jump-robust prediction

Answer: CV Kalman works well for smooth motion; jump sequences show neutral improvement vs. EMA alone (acknowledged in Section 5.7). The prediction window is essential to cover mechanical settling lag.

APPENDIX · Q&A BACKUP

Q: When and how will RPM-to-velocity calibration be completed?

Protocol designed (Section 6.4.3) — Doppler radar or high-speed camera approach

Current state

- ▶ Ballistic solver currently uses an analytic RPM-to-velocity model (theoretical relationship)
- ▶ No empirical ground-truth mapping yet: RPM commanded vs. actual ball exit speed
- ▶ This is listed as the #2 future work item after closed-loop moving-target firing

Planned calibration protocol (Section 6.4.3)

- ▶ Option A: Doppler radar gun — direct ball exit speed measurement per RPM setting
- ▶ Option B: High-speed camera (240+ fps) — pixel-tracking ball over known distance
- ▶ 50-100 shots across RPM range (0, 500, 1000, 1500, 2000 RPM per motor)
- ▶ Fit polynomial or lookup table: commanded RPM → measured exit speed (m/s)

Impact on current results

- ▶ Current ball accuracy (95.17 mm) is achieved WITHOUT empirical calibration
- ▶ Empirical calibration expected to reduce ballistic error further — accuracy will improve
- ▶ Safety clamps on RPM (firmware + Python) remain regardless of calibration state

Answer: protocol designed (Section 6.4.3), not yet executed. Doppler radar or high-speed camera approach. Current 95.17 mm accuracy is achieved without it — empirical calibration will improve results further.